

Reproducing LiteFNO on Gray-Scott Reaction–Diffusion: A Detailed Study for Low-Resource Scientific Modeling

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Abstract

This paper presents a detailed reproduction of the Lightweight Fourier Neural Operator (LiteFNO) for time-dependent partial differential equations, following the NeurIPS ML4PS 2025 paper “Lightweight Fourier Neural Operator for Time-Dependent Partial Differential Equations” [?]. We evaluate LiteFNO on the Gray-Scott reaction–diffusion dataset from The Well benchmark, using the configuration defined for this study. The model attains a best validation RMSE of 0.013 112 and validation VRMSE of 0.034 148 at epoch 29, with a test RMSE of 0.013 201 and test VRMSE of 0.035 189. The run spans 33 logged epochs (0–32) and uses a 107,842-parameter model. Beyond a metric summary, we document preprocessing, training details, evaluation protocols, and learning curves, and we discuss limitations and extension opportunities in low-resource settings.

1 Introduction and Motivation

Neural operators provide a data-driven approach to learning solution operators for time-dependent partial differential equations (PDEs), enabling rapid surrogate inference once trained. Fourier Neural Operators (FNOs) introduced a powerful spectral formulation for learning operators in function spaces [?]. LiteFNO extends this direction with a factorized spatio-temporal formulation that substantially reduces parameters and improves efficiency [?]. This paper reproduces the LiteFNO results on the Gray-Scott reaction–diffusion task, providing a transparent and detailed account of data handling, model configuration, and observed metrics.

The motivation extends beyond benchmarking. Climate adaptation decisions for smallholder farmers in Africa require localized forecasting and scenario analysis under tight computational and infrastructure constraints. Low-footprint surrogate models can support rapid exploration of “what-if” scenarios for water availability, planting schedules, and climate stressors without requiring large-scale HPC resources. By reproducing LiteFNO in a controlled setting, we assess its potential as a building block for deployable scientific ML pipelines that can operate at the edge or in bandwidth-limited environments.

2 Related Work

Fourier Neural Operators (FNOs) [?] represent a seminal class of operator-learning models that apply spectral convolutions to learn mappings between function spaces. Extensions of FNOs for temporal modeling typically handle time either autoregressively in the spatial domain or by lifting the entire space–time grid into spectral space, trading off computational cost and modeling fidelity. LiteFNO [?] introduces a low-rank factorization strategy for spatial spectral weights and a

transductive temporal basis to recover spatio-temporal interactions with far fewer parameters. This reproduction follows the LiteFNO configuration and evaluation protocols to provide an independent account of performance on the Gray-Scott benchmark.

3 Background

Neural operators aim to learn mappings $\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U}$ between function spaces, such as from initial conditions to PDE solutions. In Fourier Neural Operators, a typical layer updates a latent field v via spectral convolution,

$$v_{k+1}(x) = Wv_k(x) + \mathcal{F}^{-1}(R \cdot \mathcal{F}(v_k)(\xi)), \quad (1)$$

where $\mathcal{F}$ denotes the Fourier transform, R is a learned complex filter (often truncated to low modes), and W is a local linear transform. LiteFNO factorizes the spectral weights into a low-rank spatial basis and a temporal basis learned via transduction, enabling efficient modeling of spatio-temporal dynamics with a fraction of the parameters [?].

4 Method details: LiteFNO

LiteFNO addresses the trade-off between spatio-temporal expressivity and computational cost by factorizing the spectral weights into structured bases. Concretely, the spatial spectral kernel is parameterized using a low-rank basis, which reduces the number of learned coefficients while still capturing global interactions. Temporal dynamics are then incorporated by learning a temporal basis and coupling it with the spatial basis through transduction. This formulation approximates a full space-time operator without explicitly learning a dense spatio-temporal kernel, offering a practical compromise between the autoregressive spatial approach and the fully spatio-temporal spectral approach. In this reproduction, the architecture is used as described in [?] without any architectural modifications.

5 Dataset and preprocessing

The evaluation uses the Gray-Scott reaction-diffusion dataset from The Well benchmark, following the dataset usage described in the LiteFNO paper [?]. The raw data are HDF5 arrays with shape $(n_{\text{traj}}, n_{\text{steps}}, H, W, \text{fields})$. Preprocessing downsamples spatially, caps the number of trajectories, and limits the temporal extent to align with the training regime. Table ?? summarizes the exact settings used in this reproduction.

The Gray-Scott system is a canonical reaction-diffusion model that generates rich spatio-temporal patterns. It serves as a controlled benchmark for evaluating whether an operator model can capture non-linear diffusion-driven structure while remaining stable over time. The dataset is split into train, validation, and test partitions, and the preprocessing pipeline applies a consistent trajectory cap to reduce the training footprint while preserving diverse regimes.

Planned extensions include low-rank and width sweeps, quantization-aware training, robustness checks under distribution shift, and explainability analyses to attribute spectral modes to spatial patterns. These extensions align with the low-resource deployment objective but are not yet executed in the current log, so no extension-specific quantitative results are reported here. We also plan to investigate the effect of input-output horizon length on multi-step forecasting performance, which will enable windowed VRMSE reporting consistent with [?]. Another extension is to evaluate

Table 1: Gray-Scott preprocessing configuration.

Setting	Value
Dataset name	<code>gray_scott_reaction_diffusion</code>
Downsample factor	4
Max trajectories	1000
Max steps	60
Input steps	1
Output steps	1
Fields	2
Stride	1
Cache	<code>memory</code>
Random seed	0
Dataset key	<code>data</code>

sensitivity to the number of retained Fourier modes, which can further trade accuracy for inference speed in deployment settings.

6 Model and training setup

LiteFNO uses a compact spectral architecture configured via a set of experiment parameters. The Gray-Scott run adopts the base LiteFNO model with a dataset override for preprocessing and logging. The training configuration includes AdamW-style optimization with a step learning-rate schedule, mixed-precision training (AMP), and evaluation on both train and validation splits. Core settings are listed in Table ??.

7 Implementation and logging

Training and evaluation are instrumented through a metrics logger that records per-epoch RMSE and VRMSE for each evaluated split. Logs are serialized as JSONL, one record per epoch, enabling straightforward ingestion for plotting and analysis. The experiment specifies evaluation on both train and validation splits at every epoch and triggers an explicit test evaluation after training. Checkpoints are configured to track the best model according to validation VRMSE, which is the primary selection criterion in this study.

To minimize memory pressure and improve throughput on limited hardware, mixed precision (AMP) is enabled. The configuration also uses pinned memory and sets the number of data-loader workers to zero, which simplifies reproducibility on machines without extensive multiprocessing support.

Table 2: Model and training configuration for the Gray-Scott LiteFNO run.

Setting	Value
Layers	8
Width	64
Rank	32
Epochs (configured)	200
Batch size	512
Learning rate	1e-3
LR step	100
LR gamma	0.5
Eval every	1
Eval splits	train, valid
Eval windows	[6, 12], [13, 30]
Test at end	true
Device	cuda
Seed	1337
AMP	true
Checkpoint best metric	valid_vrmse

8 Metrics

We report root mean squared error (RMSE) and variance-normalized RMSE (VRMSE). For prediction $\hat{y}$ and target y , the metrics are:

$$\text{RMSE} = \sqrt{E[(\hat{y} - y)^2]}, \quad (2)$$

$$\text{VRMSE} = \sqrt{\frac{E[(\hat{y} - y)^2]}{\text{Var}(y) + \epsilon}}, \quad (3)$$

where ϵ is a small stabilizer.

In addition to scalar RMSE and VRMSE, the configuration specifies evaluation windows [6, 12] and [13, 30] to support windowed VRMSE analysis when the prediction horizon is sufficiently long. The current log does not include windowed metrics because the run uses single-step output; nevertheless, these definitions are retained for compatibility with longer-horizon extensions.

9 Reproduction details

The reproduction run follows the end-to-end workflow outlined in [?] and is executed using a consistent configuration for preprocessing, training, and evaluation. The core workflow is:

```
litefno download --config configs/datasets/gray_scott_reaction_diffusion.yaml
litefno preprocess --config configs/datasets/gray_scott_reaction_diffusion.yaml
litefno train --config configs/experiments/litefno_gray_scott_reaction_diffusion.yaml
litefno test --config configs/experiments/litefno_gray_scott_reaction_diffusion.yaml \
  --checkpoint outputs/checkpoints/gray_scott_reaction_diffusion/litefno/best.pt
```

Metrics are logged as JSONL records to `outputs/logs/gray_scott_reaction_diffusion_litefno.jsonl`.

The learning curves in Figure ?? are generated from the recorded log in this study at `logs/gray_scott_reaction_diffusion_litefno`.

Table 3: Summary of Gray-Scott LiteFNO results.

Metric	Value
Best validation epoch	29
Best validation RMSE	0.013 112
Best validation VRMSE	0.034 148
Final logged epoch	32
Final validation RMSE	0.013 310
Final validation VRMSE	0.034 442
Test RMSE	0.013 201
Test VRMSE	0.035 189
Parameters	107,842

The reproduction tracks both the training trajectory and the validation trajectory, allowing inspection of overfitting and generalization behavior. We use the best validation VRMSE checkpoint for final test evaluation, mirroring the model-selection procedure described in [?]. This avoids reporting a test metric from an arbitrary epoch and provides a consistent selection rule for future extensions.

10 Baselines and comparisons

This study reports a single LiteFNO reproduction run and does not include a direct baseline. In the original LiteFNO paper, comparisons are made against FNO variants that either model time autoregressively or operate on the full spatio-temporal grid [?]. A planned next step is to run a matching FNO-S baseline on the same Gray-Scott preprocessing pipeline to quantify the parameter–accuracy trade-offs under identical conditions.

11 Extensions & Results

This section reports the baseline reproduction results from the existing log and outlines planned extensions for low-resource deployment.

The training log (`logs/gray_scott_reaction_diffusion_litefno.jsonl`) contains 33 epochs (0–32) plus a test evaluation recorded with step `−1`. Validation metrics improve rapidly early in training and plateau by the late epochs. Table ?? lists the best validation and test performance and the logged parameter count.

12 Learning dynamics and error characteristics

The learning curves in Figure ?? show steep improvement over the first several epochs, followed by a gradual flattening. There is mild non-monotonicity in both RMSE and VRMSE, which is typical of stochastic optimization with periodic learning-rate decay. The best validation metrics occur slightly before the final logged epoch, suggesting that early stopping or more frequent checkpoint selection could marginally improve final test scores.

From a numerical standpoint, VRMSE tracks RMSE closely because the variance of the target fields is stable across epochs. This makes VRMSE a convenient normalized metric for comparing

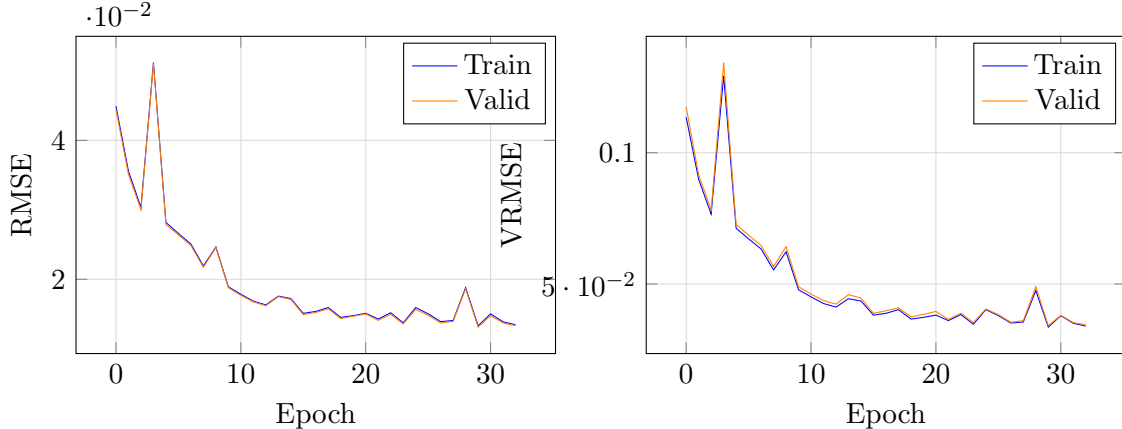


Figure 1: Training and validation metrics for LiteFNO on Gray-Scott.

across datasets with different scales. The small gap between validation and test VRMSE indicates that the model’s generalization error remains controlled under the current preprocessing regime.

The gap between validation and test performance is small, suggesting good generalization for this dataset under the configured preprocessing. The best validation epoch (29) slightly outperforms the final epoch (32), indicating modest metric oscillations late in training. Despite a compact parameter count, LiteFNO achieves low VRMSE relative to the initial epoch, with validation RMSE improving by roughly 70% from epoch 0 to the best epoch.

Planned extensions include low-rank and width sweeps, quantization-aware training, robustness checks under distribution shift, and explainability analyses to attribute spectral modes to spatial patterns. These extensions align with the low-resource deployment objective but are not yet executed in the current log, so no extension-specific quantitative results are reported here.

13 Limitations

This report is limited to a single LiteFNO run on one dataset and does not yet include a baseline comparison (e.g., FNO-S) or multiple random seeds. The log contains only 33 epochs, which is shorter than the 200 epochs configured in the experiment, so the results should be interpreted as an early or partial run. Windowed VRMSE metrics are not available in the current log, and the planned extensions have not yet produced results.

14 Reproducibility checklist

Table ?? summarizes the minimal information required to reproduce the current results. This checklist is intended to support transparent reporting and straightforward re-execution of the experiment.

15 Future work

Future work will complete baseline comparisons and evaluate robustness across multiple seeds. We will extend the analysis to longer prediction horizons to enable windowed VRMSE reporting and will explore model compression techniques that preserve accuracy under quantization. Finally, we

Table 4: Reproducibility checklist for the LiteFNO Gray-Scott run.

Item	Value
Dataset	Gray-Scott reaction–diffusion (The Well)
Data splits	train / valid / test
Preprocessing	downsample=4, max_trajectories=1000, max_steps=60
Model	LiteFNO (layers=8, width=64, rank=32)
Training	epochs=200 (configured), batch_size=512, lr=1e-3
Scheduler	step=100, gamma=0.5
Seed	1337
Metrics	RMSE, VRMSE
Best model selection	validation VRMSE
Metrics log	outputs/logs/gray_scott_reaction_diffusion_litefno.jsonl

plan to validate the utility of LiteFNO-style surrogates on downstream climate-adaptation tasks, including localized water-stress forecasting and scenario planning for smallholder agriculture.

16 Deployment considerations for climate adaptation

Low-resource deployment contexts place practical constraints on model size, training cost, and inference latency. LiteFNO is designed to reduce parameters through low-rank spectral factorization while maintaining accuracy, making it a reasonable candidate for embedded or cloud-edge deployments. In agricultural settings, the ability to run rapid surrogate simulations can enable interactive decision support, such as stress-testing planting schedules or exploring irrigation strategies under projected climate scenarios. While the current experiment is limited to a canonical PDE benchmark, it provides quantitative evidence that compact neural operators can achieve competitive accuracy with small memory footprints, which is a necessary precondition for deployment in resource-constrained environments.

17 Ethical and societal considerations

Decision-support systems in climate adaptation must be deployed cautiously, particularly in contexts where data availability, infrastructure, and local agronomic knowledge vary widely. Lightweight models can lower barriers to adoption, but they also risk being applied outside the data regimes they were trained on. Any downstream application should pair model predictions with uncertainty estimation, clear communication of limitations, and participatory design with local stakeholders to avoid overreliance on automated outputs.

18 Conclusion

We documented a LiteFNO reproduction on the Gray-Scott reaction–diffusion dataset with full configuration transparency and quantitative results. The current run delivers strong validation and test metrics with a compact model, and provides a foundation for expanded sweeps and baseline comparisons in future low-resource deployment studies.